

Dilutha Weerasinghe

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SUMMARY

Business Information Systems undergraduate with a **BSc (Hons) in Data Science with Second Class Upper Division** and currently pursuing an **MSc in Applied Artificial Intelligence**. Strong foundation in **software development, data analytics, machine learning, business analysis, and information systems**, with hands on experience building **AI powered applications, predictive models, REST APIs, databases, and full stack systems**. Proficient in **Python, SQL, TypeScript, Next.js, Flask, Spring Boot, and modern data technologies**. Strong analytical and problem solving abilities with experience translating business requirements into practical, data-driven technology solution

CORE SKILLS & TECHNICAL PROFICIENCY

- Programming Languages:** Python, R, SQL, Type Script, Java, C#
- Machine Learning & Deep Learning:** scikit learn, TensorFlow, Keras, XGBoost, Prophet, Random Forest, Regression Models
- Data Analytics & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Streamlit, Statistics
- Frameworks & Software development tools:** Flask, Laravel Blade, Git, API Integration, MVC / MVT Architecture, Spring Boot, Flutter, Django
- Data Engineering:** Data Preprocessing, Feature Engineering, Data Cleaning, Model Deployment, Advanced analytics
- Soft Skills:** Analytical Thinking, Communication, Teamwork, Autonomy, Curiosity, Problem Solving, Writing Skills
- Business & Information Systems:** Business Process Analysis, Information Systems, Database Design, Business Analytics, Systems Analysis, Software Development Lifecycle

EDUCATION QUALIFICATIONS

MSc Applied Artificial Intelligence | 2026 - PRESENT

University Of Westminster, UK (Affiliated Program - IIT, Sri Lanka) - Reading

BSc (Hons) in Business Information Systems (Special) | 2023 - PRESENT

University of Sri Jayewardenepura - Undergraduate (Current GPA: 3.89)

BSc (Hons) in Data Science | JUN 2021 - FEB 2024

Cardiff Metropolitan University, UK (Affiliated Program - ICBT Campus) - Second Class Upper Division

G.C.E. Advanced Level - Commerce Stream | JAN 2023

- *ICT, Accountancy, Business Studies* - 3 A's
- *General English* - A Pass

G.C.E. Ordinary Level | DEC 2019

- *9 A's including Mathematics, Science, ICT, and English*

EXPERIENCE & PROJECTS

PneumoScan AI - Deep Learning Based Pneumonia Detection from Chest X-rays

- Designed and implemented a **DenseNet121, GradCAM Explainability** to classify pneumonia from medical images.
- Performed **data preprocessing** and **feature engineering** using **TensorFlow, Keras, and scikit learn**.
- Built an interactive **Next.js, FastAPI full stack** for model inference and results visualization.
- Utilized **Git** for version control and model tracking.
- https://github.com/dilutha/Pneumonia_X-tray_Detection

DengueSense - Hybrid AI Decision Support System

- Developed a **hybrid AI decision-support system** for early dengue outbreak prediction and risk assessment in Sri Lanka.
- Integrated **LightGBM, XGBoost, Prophet, SHAP, and a Forward Chaining Expert System** for prediction, forecasting, explainability, and recommendations.
- Built an end to end architecture using **Next.js, FastAPI, Supabase PostgreSQL, and Python based ML models**.
- Implemented a **Decision Fusion Engine** combining model outputs and expert system reasoning to generate risk levels and recommendations.
- <https://github.com/dilutha/DengueSense>

RidePulse - AI Driven Public Transport Intelligence System

- Built a **conductor based digital ticketing system** with real time revenue and GPS data capture.
- Integrated **ML models** for demand forecasting and route performance analytics using Python.
- Developed scalable backend services using **Spring Boot and PostgreSQL** and Frontend using **Flutter**.
- <https://github.com/dilutha/RidePulse>

CareerLens AI - AI Powered Career Intelligence Platform

- Developed an **AI powered career platform** using **Next.js, TypeScript, Supabase, and Google Gemini** for personalized job matching.
- Implemented **job requirement extraction, resume analysis, CV tailoring, cover letter generation, ATS scoring, and application tracking.**
- Integrated **Google Gemini SDK** and **Supabase PostgreSQL** to support AI driven workflows and application data management.
- Designed a fullstack application using **Next.js App Router and Tailwind CSS.**
- <https://github.com/dilutha/CareerLens-AI>

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **President, S@IT** (Student Association of IT), University of Sri Jayewardenepura for year 2026
- **Organizing Committee Member**, D'Hack & Sparsha 2025, 2026
- **School Senior Prefect**, and active member of **Cricket, Chess, and Debate Teams.**

ACHIEVEMENTS

- **Google Advanced Data Analytics Certificate** (Coursera)
- **District Rank 84** in G.C.E. Advanced Level (Commerce Stream)
- **9A's** at G.C.E. Ordinary Level
- Completed several **AI and ML projects** demonstrating applied data science proficiency.

REFERENCES

Ms. Upeksha Samarasinghe

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